Reg. No. | | | | | | | | | | | [12CS73]

SEVENTH SEMESTER B.E EXAMINATION DECEMBER –2015

DIGITAL IMAGE PROCESSING

Time:3 Hours **Maximum Marks:70**

Instruction: Answer any **FIVE** full Questions.

1. a. Briefly explain the process of sampling and quantization. **07**

b. Explain some basic relationships between pixels with examples. **07**

2. a. Explain the basic intensity transformation function with diagram. **06**

b. A 8 level image of size 64×64 has the density distribution as shown in table. Implement histogram equalization and plot the same. **08**

r_k	0	1	2	3	4	5	6	7
n_k	790	1023	850	656	329	245	122	81

3. a. Explain the following: **06**

- **i.** Ideal low pass filters
- **ii.** Butter worm low pass filters
- **iii.** Gaussian low pass filters

b. Describe the process of homomorphic filtering with derivation. **08**

4. a. Define optimal thresholding. Explain with an algorithm for optimal threshold selection. **07**

b. Explain the algorithm procedure for Edge relaxation. **07**

5. a. Explain the following: **09**

- **i.** Region merging
- **ii.** Region splitting
- **iii.** Splitting and merging

b. Explain watershed segmentation with one-dimensional example. **5**

6. a. Explain various image compression models. **10**

b. Describe Lossy Predictive coding with neat diagram. **04**

7. a. Explain Gray-scale dilation and erosion in morphological operation. **07**

b. Describe briefly Morphological segmentation for watershed segmentation algorithm. **07**

8. a. Write an algorithm for 4-neighborhood and 8-neighborhood region identification with example. **07**

b. Explain Region Convex Hull algorithm. **07**

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